

Expansion of Body Adaptation Areas for Prism Adaptation Therapy Using Virtual Reality: Head Orientation Experiments on Normal Subjects

Takashi Ishida^a, Student Member

Hiroki Higa, Member

We developed a system to conduct **posture-measurable virtual prism adaptation therapy** using an inside-out tracking stand-alone virtual reality system. This system enables the construction of a unilateral spatial neglect (USN) treatment room that is closed off from external stimuli from all surroundings in a hospital or home. Furthermore, it expands the prism adaptation body adaptive site, which was **previously limited to the hand, to include the whole body using a symbolic body**. This expansion allows for utilization of the new prism adaptation therapy in which giving postural feedback and the head is spontaneously turned to the neglected side. This prism adaptation is expected to improve the posture of patients with USN and turn the field of vision itself toward the neglected space. © 2022 Institute of Electrical Engineers of Japan. Published by Wiley Periodicals LLC.

Keywords: unilateral spatial neglect; prism adaptation; virtual reality

Received 25 April 2022; Revised 28 July 2022

1. Introduction

Unilateral spatial neglect (USN) is a condition in which, after stroke, attention to the space contralateral to the affected cerebral hemisphere is diminished, and attention to the space on the affected side is disproportionately high. Patients with USN may ignore people located on their neglected side of the vision field or become lost without noticing the aisle on their neglected side (Fig. 1). As a result, patients with symptoms of USN may experience significant difficulties in their daily lives, making it very difficult to reintegrate into society after incurring a stroke. USN is known to be more severe in right hemisphere injuries than in left hemisphere injuries, and the prognosis is worse in left hemisphere injuries [1,2].

The heads of patients with USN always face the cognitive side because they try to position their bodies in the cognitive space. Here, when the therapist attempts to turn the USN patient's head toward the neglected side manually, the USN patient responds by turning the head toward the cognitive side more than before the modification. This abnormal head orientation results in a vision bias toward the cognitive side and makes it more difficult for the patient to attend to the neglected space [3,4].

Numerous USN rehabilitation methods have been proposed, such as visual scanning training, visual feedback, and auditory feedback [1,4]. Most of these methods improve the training activity's outcomes, but not those of other activities. In other words, most conventional USN rehabilitation methods have little evidence of generalization effects. The effect of rehabilitation is limited to the specific activity in which the patient was trained and did not spill over to activities of daily living (ADLs); therefore, the

significance of conducting the rehabilitation is very limited. The lack of generalization effects is the reason why USN rehabilitation has been considered difficult. Recently, prism adaptation (PA) has been investigated as a promising treatment method for USN, to reintegrate visual and somatosensory perception by having subjects wear prism glasses and repeatedly perform a task to reach a target object with their hands. Rossetti *et al.* reported alleviation of USN symptoms after completing PA [5]. Figure 2 shows the figure copying task before and after PA reported by Rossetti *et al.* Before PA, the left space was not copied due to left USN symptoms, but after PA, the area copied was expanded to the left. (The details and analysis of the PA task and the principle underlying the alleviation of USN by PA are described later in Section 2). This research reported the generalization effects of PA and highlighted PA as a promising method for USN rehabilitation. Since then, PA has been the subject of many studies. Later studies have shown that PA also exhibits generalization effects on ADLs [6], which has been favorably received. PA has been actively studied for synergistic effects in combination with conventional USN rehabilitation [1]. The aforementioned research makes the principles, implementation, and expansion of the adaptive range in PA a significant topic in medicine, psychology, and engineering.

Finally, we describe the development status of VR systems that conducted PA in the previous study. In recent years, Kim *et al.* [7] and Cho *et al.* [8] conducted experiments on healthy subjects and reported that PA could be realized using VR instead of prism glasses (hereafter referred to as VPA). However, both studies only replaced prism glasses with VR. In other words, no attempt to measure or improve the posture of patients with USN was conducted. Both studies used PC-connected VR with inside-out tracking, which is expensive in terms of both equipment and operational costs, as it requires a dedicated installation site. It is unrealistic to secure an expensive dedicated device and its installation site only for patients with USN in clinical practice. In addition, treatment of higher brain dysfunction such as USN

^a Correspondence to: Takashi Ishida. E-mail: k208672@eve.u-ryukyuu.ac.jp

Graduate School of Engineering and Science, University of the Ryukyus, 1 Senbaru, Nishihara, Nakagami District Okinawa, 903-0213, Japan

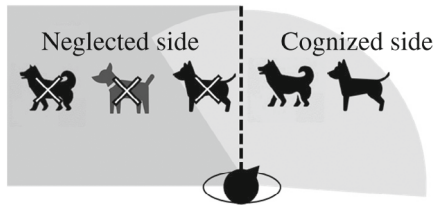


Fig. 1. Neglected side and posture of USN patients

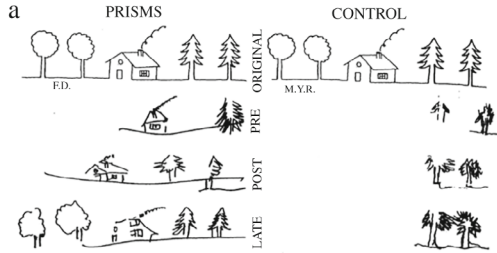


Fig. 2. Alleviation of USN with PA, quoted from Rosetti *et al.* 1998 [5]

generally requires time in the order of years. It is desirable to develop a portable device that enables training at home after discharge from the hospital.

Based on the results of these investigations, the objectives of this study were defined as follows. The first objective was to reduce costs and add a posture measurement function. In other words, the first objective was to develop an inexpensive and portable VPA system that measure and record the posture of patients with USN. The second objective was to analyze the effect of PA on posture. Using the posture measurement function in the developed system, we aimed to clarify the effect of PA on the head orientation, which has remained unclear. The third objective was to expand the range of applications of VPA. In the past, PA and VPA have been applied only to the hand, but we expanded the range of application to the head by using the additional posture measurement function. This expansion enables patients with USN to spontaneously turn their heads toward the neglected side.

2. Analysis of Conventional PA Therapy

This section describes the conventional PA task and analyzes the USN alleviation principle with this task. (From this point onward, the left-sided USN, which is the more common condition and has a poorer prognosis, will be discussed).

2.1. How to perform PA The conventional PA implementation method is described here. First, the subject is placed in an upright sitting position on a chair. The blindfold plate is placed under the chin to block the view so that the subject cannot see the hand in the start position. Next, the subject is given a task to quickly extend their upper limb and point to a target in front of them (Fig. 3). Because of the blindfold plate, the subject was given only knowledge of the outcome, i.e., the error in the distance between the target and the finger at the end of the exercise. As a result, the subject conducts this task with open-loop control. The above task is performed several times: first without prism glasses, then with prism glasses, and then without prism glasses for a second time.

2.2. How to alleviate USN symptoms

Here, we analyze the principle by which PA alleviates USN symptoms. Figure 3 illustrates the details of this principle. In the figure, the circle represents the target, and the hand icon represents the hand of the USN patient. Prism glasses deflect visibility to the right. Here, the black fill represents the real image, and the white fill represents the imaginary image through the prism glasses. Without prism glasses, the USN patient sees only the real image, and with prism glasses, the USN patient sees only the imaginary image deflected to the right by the prism glasses.

i Normal, without prism glasses.

USN patient points exactly at the real target.

ii Normal, with prism glasses.

The patient with USN will see the imaginary target instead of the real target when wearing prism glasses. Therefore, the patient points at the imaginary target. However, since the patient with USN sees the indicated hand through the prism glasses, the patient sees the imaginary hand instead of the real hand. Therefore, the subject perceives that the hand is rightward shifted from the target.

iii Prism adapted, with prism glasses.

The patient with USN learns to align their imaginary hand with the imaginary target by reaching their real hand to the left of the imaginary target. This phenomenon is called PA.

iv Prism adapted, with prism glasses.

After task iii is sufficiently completed, the target is presented so that the imaginary image is positioned in the cognitive space close to the neglected space. The USN patient usually cannot recognize the neglected space, so they do not intend to reach their hand into the neglected space. However, the patient can reach the neglected space to align their imaginary hand with the imaginary target because he/she has already adapted to the prism by the task in part iii.

v Normal, without prism glasses.

Even if the prism glasses are removed, the patient continues to point to the left of the real image for a brief period (varying in length among conditions) because he/she has still adapted to the prism.

In an experiment conducted by Rossetti *et al.* on patients with left USN [5], symptoms were alleviated when patients were fitted with prism glasses that deviated their vision to the right (cognized side), and PA was performed to reach the left (neglected side) as described in this section. This result suggests that hand-seeking behavior in the neglected side is a factor in alleviating USN symptoms. In other words, patients with USN perform upper limb activities to the neglected side more than the normal state in processes iii–v, and this treatment was found to improve spatial cognition on the neglected side.

3. Design of Virtual PA System

This section describes the outline of the VPA system developed in this study.

3.1. System configuration Figure 4 shows the block diagram of the proposed system. The hardware was developed using Oculus Quest2 as a VR headset (Oculus, Menlo Park,

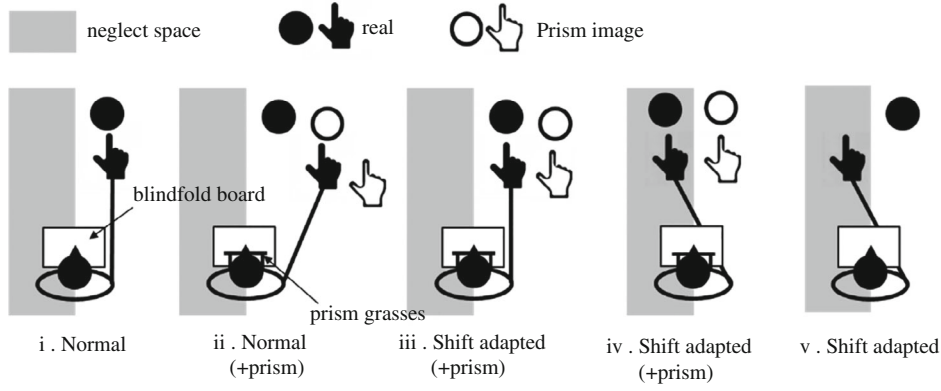


Fig. 3. Principle of prism adaptation

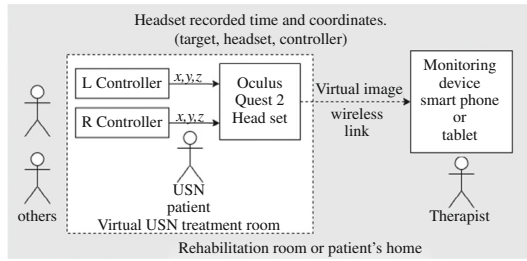


Fig. 4. Block diagram of virtual prism adaptation system

CA, USA) and Android smartphone version 6.01 (Android, Mountain View, CA, USA) as a monitoring device. The software was developed using Unity 2021.2.2 (Unity Technologies, San Francisco, CA, USA) and Oculus Integration SDK version 38.0. The use of a stand-alone VR with inside-out tracking made it possible to construct a dedicated virtual USN treatment room while in the same space as others. The VR is connected to the monitoring device via a wireless link, allowing the therapist to observe both the virtual view of the USN patient and their actual physical movements without entering the virtual space.

3.2. Body measurement and target position First, body measurements were taken to determine spatial coordinates and target positions.

The subjects positioned the controller at the left and right greater trochanter, respectively, in the end-seated posture on a chair and pulled the left and right trigger buttons of the controller at the same time. The coordinates of the left and right controllers at this time were recorded as the coordinates of the left and right greater trochanter GT_L and GT_R , respectively (Fig. 5(a)). Next, the subject pulled the trigger button with the dominant arm in a posture of 90° shoulder flexion, 0° abduction, and full elbow extension (Fig. 5(b)). At this time, the upper limb length l_a and shoulder joint height h_s were measured and recorded from the coordinates of the controller held in the dominant hand, and the eye height h_e from the camera coordinates (Fig. 5(b)). The origin and axis were then determined from the measurements: The midpoint between the GT_L and GT_R was taken as the origin, and the line connecting GT_L and GT_R was taken as the x-axis and the perpendicular bisector of this line as the y-axis (Fig. 5(a)). The z-axis was defined to be perpendicular to the xy plane through the origin (Fig. 5(b)). Targets appeared at 10° locations at 30° angles on either side of the y-axis in 6° increments

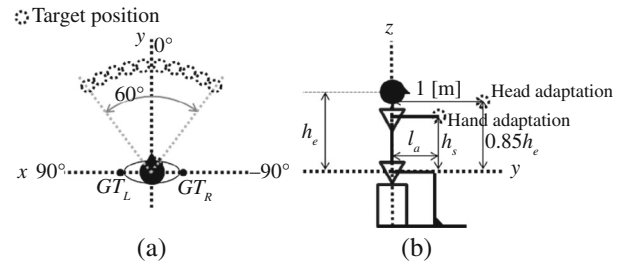


Fig. 5. Body measurements and target positions [Correction added on 8 December 2022, after first online publication: Figure image has been corrected.]

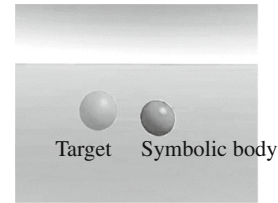


Fig. 6. Symbolic body and target in display

(Fig. 5(a)). The height and distance of the targets were calculated as shown in Fig. 5(b) according to individual differences in the body.

3.3. Symbolic body The user interface in the developed system is described below. Most USN patients have concomitant attention disorders and face difficulty understanding complex instructions [2,4]. Pre-tests using avatars that imitated hands and heads have shown that it is difficult for USN patients to know the position error. The shape of the hand, head, and targets are superfluous information in a USN patient's spatial positioning task and can distract their attention from the spatial problem. Therefore, spheres are used instead of the usual avatars of hands and heads (right sphere in Fig. 6), called a symbolic body (SB). The target was a translucent sphere (left sphere in Fig. 6). The spherical shape was adopted because it is the best symbol for positioning tasks considering its specific size and ease of determining the center. By adopting this user interface, the positional error between the target and the SB can be judged easily, even by a patient with an attention disorder. The subject performed the task of controlling the SB to overlap it with the target.

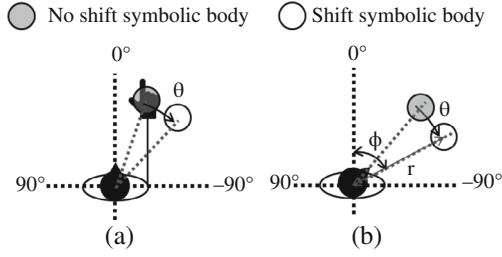


Fig. 7. Virtual prism adaptation tasks. (a) Hand adaptation. (b) Head adaptation

3.4. Virtual PA tasks Two types of VPA tasks were created and implemented: Hand adaptation was a VR implementation of a conventional PA, and head adaptation was a newly proposed task in this study in which PA was generated for the head.

3.4.1. Hand adaptation (HaA) One of the 10 target placement locations shown in Fig. 5(a) was randomly selected, and the target was displayed. The subject moved the controller in his hand to the target location and pulled the trigger (Fig. 7(a)). The SB was displayed only while the trigger was pulled (Fig. 6); the SB was displayed at the controller position for the no-shift condition and at a position rotated by shift angle θ around the origin from the controller position for shift condition (Fig. 7(a)). Let the x and y coordinates of the controller be con_x and con_y , and $R(\theta)$ be the rotation matrix, then the x and y coordinates of SB on the no-shift condition, NS_x and NS_y , and the x and y coordinates of SB on the shift condition, S_x and S_y , are expressed as follows:

$$\begin{pmatrix} NS_x \\ NS_y \end{pmatrix} = \begin{pmatrix} con_x \\ con_y \end{pmatrix} \quad (1)$$

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \quad (2)$$

$$\begin{pmatrix} S_x \\ S_y \end{pmatrix} = R(\theta) \begin{pmatrix} NS_x \\ NS_y \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} con_x \\ con_y \end{pmatrix} \quad (3)$$

The amount of shift angle θ was defined as -15° , which is considered to have the most significant effect from the report by Carine *et al.* [9]. When the trigger was released, the SB and the target disappeared. This sequence of actions was considered as one task. The task was performed 20 times as a set; sets 1–3 were the no-shift condition, sets 4–6 were the shift condition, and sets 7–9 were the no-shift condition. Subjects performed sets 1–9 consecutively.

3.4.2. Head Adaptation (HeA) One of the 10 target placement locations shown in Fig. 5(a) was randomly selected, and the target was displayed. The subject faced the direction of the target location and pulled the trigger (Fig. 7(b)). The SB was displayed at the intersection of the central axis of the camera and a circle with a radius of 1 m centered at the origin for the no-shift condition and at a position rotated by shift angle θ around the origin from this position for shift condition (Fig. 7(b)).

Let $R(\theta)$ be the rotation matrix, then the x and y coordinates of SB on the no-shift condition, NS_x and NS_y , and the x and y

coordinates of SB on the shift condition, S_x and S_y , are expressed as follows:

$$\begin{pmatrix} NS_x \\ NS_y \end{pmatrix} = r \begin{pmatrix} \sin \theta \\ \cos \theta \end{pmatrix} \quad (4)$$

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \quad (5)$$

$$\begin{pmatrix} S_x \\ S_y \end{pmatrix} = R(\theta) \begin{pmatrix} NS_x \\ NS_y \end{pmatrix} = r \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \sin \theta \\ \cos \theta \end{pmatrix} \quad (6)$$

The amount of shift angle θ was defined as -15° , same as the HaA and $r = 1$. When the trigger was released, the SB and target disappeared. This sequence of action was considered one task. The number of tasks and the shift conditions for each set was the same as for Head Adaptation.

4. Experiment

4.1. Subjects and methods The development system includes newly proposed PA tasks; therefore, it is unknown whether the newly proposed PA will work as expected and the number of task trials required for PA. In addition, it would be unethical to immediately adopt the newly proposed system to patients with USN. Accordingly, in this experiment, we verified the operation of the proposed system and analyzed the adaptation process of the HaA and HeA in healthy subjects. Patients with left USN operate with their right upper limb because they have left-sided paralysis. Considering this limitation, the selection criteria for subjects were healthy adults with no history of cerebrovascular disease, and the exclusion criteria were those who were less than 80% on the Edinburgh Handedness Inventory and those who had motor problems in using VR. Fifteen subjects were selected, two of whom were excluded. Subjects performed both HaA and HeA. In accordance with the Declaration of Helsinki, this experiment was conducted after obtaining approval from the ethical review committee of Okinawa College of Rehabilitation and Welfare. Informed consent was obtained from the subjects.

4.2. Measurement indices Two indices were measured: the controller deflection angle, which is the deflection angle between the target and the controller in the horizontal plane (Fig. 8(a)), and the camera deflection angle, which is the deflection angle between the target and the camera nose (Fig. 8(b)).

4.3. Statistical processing and results The average time required for HeA and HaA was 311.9 ± 109.6 (sec) and 336.5 ± 183.9 (sec), respectively. The statistical process of HaA is described as follows: First, the mean values of controller deflection

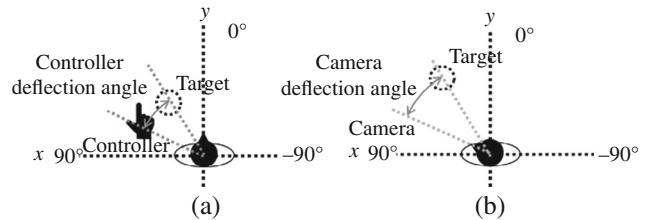


Fig. 8. Measurement indices [Correction added on 8 December 2022, after first online publication: Figure image has been corrected.]

angle and camera deflection angle for each subject in each set were calculated, and box plots (interquartile range) were created for each (Fig. 10(a, b)). Second, Steel-type multiple contrast tests were performed for sets 3 and 4–9, with set 3 considered as the baseline for the controller deflection angle and camera deflection angle (Fig. 10(a, b)). The statistical process of HeA was conducted similar to the statistical process of HaA (Fig. 10(c)). In addition, scatter plots were created for the controller deflection angle of HaA and the Camera deflection angle of HeA, and Spearman's rank correlation was calculated (Fig. 11). R version 4.03 was used for statistical processing.

5. Discussion

First, we discuss the primary objectives of reducing costs and adding a posture measurement function. Prior to designing the system, we interviewed a total of 25 rehabilitation professionals, including 12 physical therapists (PT), 8 occupational therapists (OT), and 5 speech therapists (ST). The following issues were identified as a result of this survey. Patients with USN tend to be easily distracted and require a private room during rehabilitation where they are not exposed to external stimuli. However, it is economically burdensome to have a dedicated training room to which a therapist has been assigned for patients with USN. In addition, the recovery period for USN is several years, and it is difficult to provide long-term rehabilitation for patients with USN at medical institutions under the current healthcare system [1,2]. The VPA in the previous study, which was developed with outside-in tracking PC-connected VR, did not address these problems because it required a dedicated equipment installation site [7,8]. In this study, by using a stand-alone VR with inside-out tracking, we virtually constructed a dedicated training room and training equipment. Developing stand-alone VR has not only made it possible to train in an independent virtual training room while in the same room with others in a medical facility, but also made it possible to take the VR equipment home and continue training after discharge from the hospital. Table I shows the results of the questionnaire survey conducted after the system was developed. Most of the issues raised by conventional PA and VPA have been solved, but clinical trials are necessary for validating this inference further.

Next, we discuss the added new function, posture measurement. Inside-out tracking VR can obtain only the relative position

information between the headset and the controller due to its principle. The inside-out tracking system sets the camera's central axis at 0° in a software startup. This is not a problem for healthy subjects because they can adjust their posture accordingly (Fig. 9(a)). However, most USN patients have postural problems that are difficult to correct [3,4]. Therefore, the posture at the reference angle 0° in a horizontal plane varies from patient to patient (Fig. 9(b)). In other words, a conventional VR system is unable to obtain comparable postural data with other patients, which is essential for analyzing and treating USN symptoms. To solve this problem, we focused on the fact that the pelvis is immobile because PA is performed with the patient sitting upright in a chair and measured the positions of both the greater trochanters. The midpoint of GT_R and GT_L was set as origin, and the perpendicular bisector to the line segment connecting the points GT_R and GT_L was set at 0° . By constructing this coordinate system based on the trunk, the developed system enables the measurement of posture data of USN patients, such as the position and orientation of the hands and head, and this posture data enable a comparative analysis of other patients during PA (Fig. 9(c)).

Furthermore, we discuss the analysis of the effect of PA on posture, which was the second objective of this study. From this perspective, consider the HaA in which the VPA realizes the conventional PA for the hand. Figure 10(a) shows that significant differences from the baseline third sets were confirmed in the fourth to seventh sets, and the mean value reached 14.00° , which is almost that of the adaptation goal, in the sixth set in the controller deflection angle. This result shows that the search behavior of the hand to the left side (neglected side) of the target occurred during the four sets. Figure 10(b) shows that no significant difference was found between the baseline third sets and fourth to ninth sets, and the mean value of the camera deflection angle in all sets was -0.5° to the right (cognitive side) of the target in the camera deflection angle. This result occurs for the following reason. All subjects were right-handed in order to make the conditions the same as those of patients with USN (the left side was the healthy side). Therefore, the subject's head turns to the right side to look in the direction of the right hand. These results indicate that the conventional PA promotes manipulation in the neglected space for the hand but also promotes manipulation in the cognitive space for the head. In other words, conventional PA can have a detrimental effect on posture, especially head orientation.

Table I. Evaluation of the developed system design by rehabilitation specialists

This survey was conducted for 12 PT, 8 OT, and 5 ST.		
Question	Can solve (%)	Cannot solve (%)
Q1 Can the proposed system solve the economic problem of dedicated training rooms and dedicated therapists or not?	96.0	4.0
Q2 Can the proposed system solve the problem of continuing rehabilitation of patients after discharge or not?	92.0	8.0
↓		
This survey was conducted for 3 PT who answered in Q1 and Q2 that the issue had not been solved.		
Question	Answer	(%)
Q3 Please describe why the developed system cannot solve Q1 and Q2.	Theoretically feasible but needs to be tested operationally in a clinical setting.	100.0

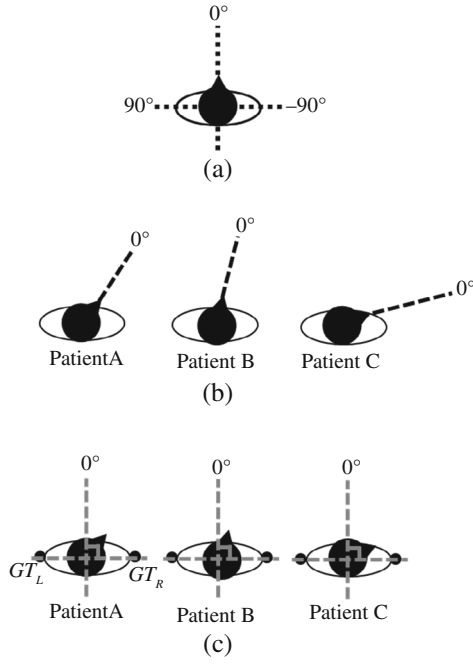


Fig. 9. Posture measurement with inside-out tracking. (a) Healthy subject. (b) Conventional inside-out tracking VR. (c) Developed system

Next, we discuss the third objective, which is to extend the adaptive range of VPA. Based on the results of HaA experiment, we developed a new HeA in order to generate the effect of turning the head to the neglected side. Figure 10(c) shows that significant differences from the baseline third sets were confirmed in the fourth to seventh sets, and the mean value reached 14.58° , which is almost the adaptation goal, in the sixth set in the camera deflection angle. These results indicate that the HeA can promote turning the head to the neglected space spontaneously. The HeA can not only correct the posture by turning the head toward the neglected space but also turn the field of vision itself toward the neglected space. HeA is expected to enhance the perception of the neglected space in patients with USN. The target of conventional PA has been limited to the hand because the hand is only the body in a visible space that can be recognized through prism glasses, which is also true for VPA [7,8]. The developed system measured posture and was represented symbolically by SB. This function made it possible that the head, which is not in the visible space, be visible in the visible space. SB enables the head to the PA phenomenon to occur. Although HeA is our newly proposed treatment for the head, the shift angle was fixed at -15° in this experiment. Therefore, it is necessary to verify the influence of the treatment effect by changing the shift angle in the future.

Finally, we discuss the HaA and HeA adaptation processes. The camera deflection angle result in HeA (Fig. 10(a)) is precisely the same as the hand deflection angle result in HaA (Fig. 10(c)). In addition, the correlation coefficient between the two is 0.70 (Fig. 11), which is very high. These results suggest that the learning processes of the hand of HaA and the head of HeA are similar and indicate that the number of trials required for adaptation does not change significantly even if the adapted body part is changed. Thus, shift adaptation needs three sets after the shift condition and has a residual effect during one set. In this

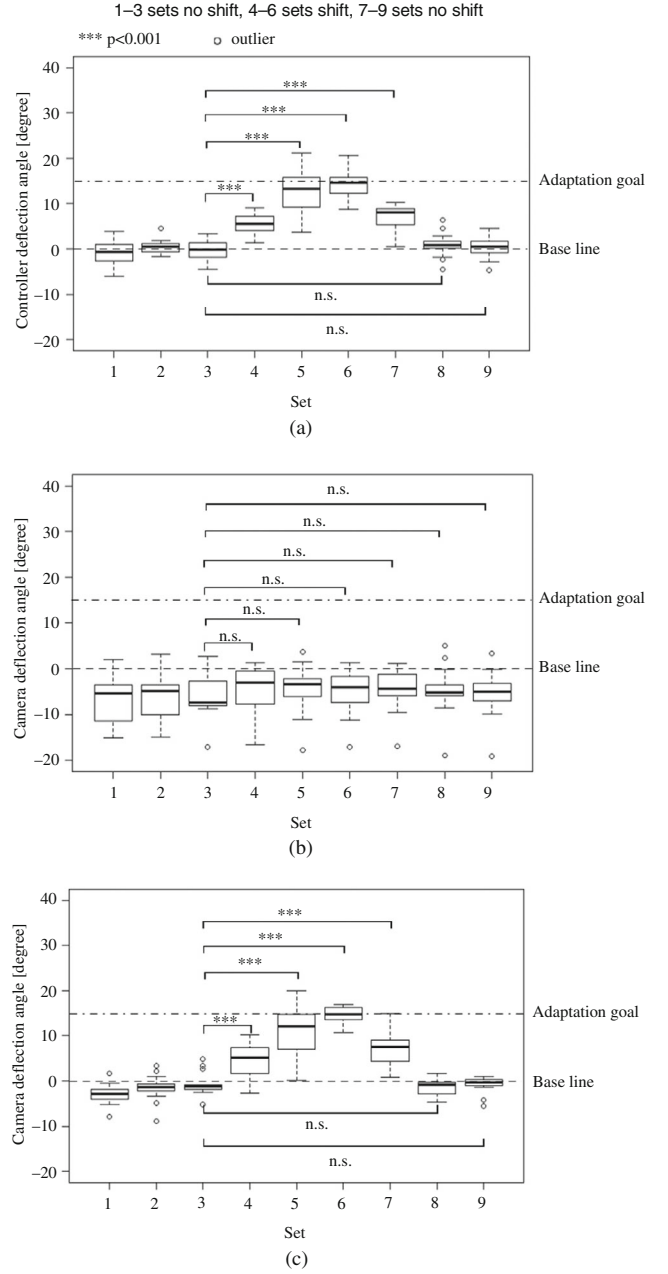


Fig. 10. Shift adaptation results. (a) Controller deflection angle on hand adaptation. (b) Camera deflection angle on hand adaptation. (c) Camera deflection angle on head adaptation [Correction added on 8 December 2022, after first online publication: ‘1–3 sets no shift, 4–6 sets shift, 7–9 sets no shift’ were added in the upper part of the figure and ‘head’ in the figure caption (a) has been amended to ‘hand’.]

period, the subject’s body reaches into space on the neglected side rather than the presented target.

6. Conclusion

This newly developed system makes it possible to construct a virtual USN treatment environment regardless of location, such as at a medical institution or home, with no financial burden. The

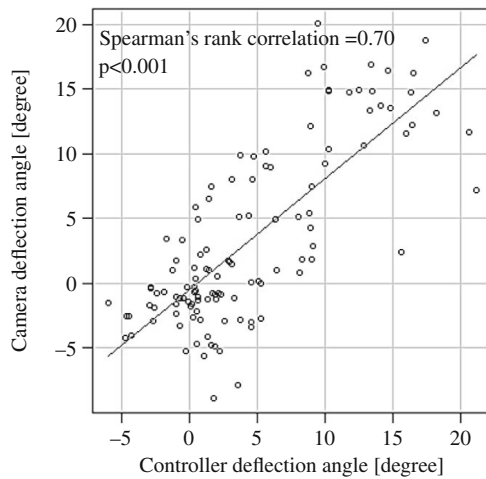


Fig. 11. Controller and camera deflection angle [Correction added on 8 December 2022, after first online publication: Figure image has been corrected.]

developed system also has an unprecedented posture measurement function. The analysis by the posture measurement function revealed that conventional PA promotes manipulation in the neglected space for the hand but promotes manipulation in the cognized space for the head. Therefore, we developed a new HeA in order to generate the effect of turning the head to the neglected side. This HeA not only corrects posture by turning the head toward the neglected space but also turns the field of vision itself toward the neglected space. From these results, it can be said that the developed system not only constructs a virtual treatment environment at a low cost but also succeeds in expanding the range of application of PA to other body parts. Patients with USN also have other posture problems with a tilt of the trunk to the cognitive side [3,4]. VPA can expand to the trunk in the future. We will verify the effectiveness of the developed system in clinical practice using USN cases.

References

- (1) Gammeri R, Iacono C, Ricci R, Salatino A. Unilateral spatial neglect after stroke: Current insights. *Neuropsychiatric Disease and Treatment* 2020; **16**:131–152.
- (2) Katz N, Hartman-Maeir A, Ring H, Soroker N. Functional disability and rehabilitation outcome in right hemisphere damaged patients with and without unilateral spatial neglect. *Archives of Physical Medicine and Rehabilitation* 1999; **80**(4):379–384.
- (3) Chen P, Motisi SE, Cording C, Ward I, Jasey NN. Impact of eliminating visual input on sitting posture and head position in a

patient with spatial neglect following cerebral hemorrhage: A case report. *Physiotherapy Theory and Practice* 2021; **37**(7):852–861.

- (4) Shinsha N, Ishigami S. Rehabilitation approach to patients with unilateral spatial neglect. *Topics in Stroke Rehabilitation* 1999; **6**(1): 1–14.
- (5) Rossetti Y, Rode G, Pisella L, Farne A, Li L, Boisson D, Perenin M-T. Prism adaptation to a rightward optical deviation rehabilitates left hemispatial neglect. *Nature* 1998; **395**(6698): 166–169.
- (6) Mizuno K, Tsujimoto K, Tsuji T. Effect of prism adaptation therapy on the activities of daily living and awareness for spatial neglect: A secondary analysis of the randomized, controlled trial. *Brain Sciences* 2021; **11**(3):347–357.
- (7) Won-Seok Kim, Nam-Jong Paik, and Sungmin Cho: Development and validation of virtual prism adaptation therapy. *International Conference on Virtual Rehabilitation* 2017.
- (8) Cho S, Kim W-S, Park SH, Park J, Paik N-J. Virtual prism adaptation therapy: Protocol for validation in healthy adults. *Journal of Visualized Experiments* 2020; **12**:156–173.
- (9) Michel C, Cruz R. Prism adaptation power on spatial cognition: Adaptation to different optical deviations in healthy individuals. *Neuroscience Letters* 2015; **17**:145–149.

Takashi Ishida (Student member) received a Master's degree



in mechanical engineering from the Tokyo Institute of Technology, Tokyo, Japan, in 1998. He is currently a lecturer at the Department of Physical Therapy, Okinawa College of Rehabilitation and Welfare and a doctoral student at the Graduate School of Science and Technology, University of the Ryukyus.

His research interests include rehabilitation engineering, with a particular focus on motor learning using virtual reality.

Hiroki Higa (Member) received the B.E. and M.E. degrees



in Electronics and Information Engineering from University of the Ryukyus, Okinawa Japan in 1992 and 1994, respectively, and the Dr. Eng. degree in Electronics Engineering from Tohoku University, Sendai, Japan in 1997. Since 1997, he has been with the Department of Electric and Electronics Engineering, University of the Ryukyus, where

he is currently a Professor. From 2008 to 2009, and 2016, he was the visiting researcher at Old Dominion University, VA, USA. His research interests include robotics and assistive device for people with disabilities.